

2(a)(i)	distance in a specified direction (from a point)	B1
2(a)(ii)	change in velocity / time (taken)	B1
2(b)(i)	constant velocity so no resultant force	B1
	no resultant force so in equilibrium	B1
2(b)(ii)	(difference in height =) $47 \times 2.8 \times 60 \times \sin 24^\circ = 3200 \text{ m}$	A1

Question	Answer	Marks
2(b)(iii)	1 $(\Delta)E = mg(\Delta)h$ $= 85 \times 9.81 \times 3200$	C1
	$= 2.7 \times 10^6 \text{ J}$	A1
	2 <u>In terms of energy:</u> work done $= 2.7 \times 10^6 \text{ J}$ force $= 2.7 \times 10^6 / (47 \times 2.8 \times 60)$	C1
	$= 340 \text{ N}$	A1
	<u>In terms of forces:</u> component of weight along path = force due to air resistance force $= 85 \times 9.81 \times \sin 24^\circ$	(C1)
	$= 340 \text{ N}$	(A1)
2(b)(iv)	$(\Delta)p = \rho g(\Delta)h$ $(92 - 63) \times 10^3 = \rho \times 9.81 \times 3200$	C1
	$\rho = 0.92 \text{ kg m}^{-3}$	A1

Question	Answer	Marks
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